

INTERVAL ESTIMATION

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HOW CONFIDENT DO YOU FEEL? LET'S TEST IT

In Article No.18, we introduced parameter estimation and explored different methods for point estimates of any unknown parameters. Whilst point estimation can provide a single best guess for a parameter value, it struggles to convey information about the uncertainty associated with the estimate. In practice, different samples from the same population will generally produce different estimates, making it important to quantify the degree of uncertainty surrounding our conclusions. Interval estimation uses the sample data to calculate an interval of possible values of an unknown population parameter with a certain degree of reliability.

The most prevalent forms of interval estimation are confidence intervals, where the random interval contains the unknown parameter with a specified probability.

FREQUENT DISTRIBUTIONS FOR CONFIDENCE INTERVALS

To calculate confidence intervals, we first need to introduce some distributions that we frequently use. That is the χ^2 , t , and F distributions.

2.1 The χ^2 Distribution

If $X_1, X_2, \dots, X_n \sim N(0, 1)$ are independent, then $\sum_{i=1}^n X_i^2 \sim \chi_n^2$. That is, the square of a standard normal random variable is χ_1^2 . Note that χ_n^2 is a gamma random variable, where:

$$f_{\chi_n^2}(x) = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}, \quad x \geq 0. \quad (1)$$

2.2 The t Distribution

If $Z \sim N(0, 1)$ and $U \sim \chi_n^2$ are independent, then the distribution of $\frac{Z}{\sqrt{\frac{U}{n}}}$ is called the t distribution with n degrees of freedom, with density:

$$f(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi} \Gamma(\frac{n}{2})} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}. \quad (2)$$

Note that the density of the t distribution is symmetric about zero, and approaches the standard normal density as the number of degrees of freedom converges to ∞ , i.e., $d.f. \rightarrow \infty$.

2.3 The F Distribution

If U and V are two independent χ^2 random variables with degrees of freedom m and n respectively, then the distribution of $W = \frac{\frac{U}{m}}{\frac{V}{n}}$ is called the F distribution with m and n degrees of freedom and is denoted by $F_{m,n}$. The density of an $F_{m,n}$ random variable is:

$$f(w) = \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2}) \Gamma(\frac{n}{2})} \left(\frac{m}{n}\right)^{\frac{m}{2}} w^{\frac{m}{2}-1} \left(1 + \frac{m}{n}w\right)^{-\frac{m+n}{2}}, \quad w \geq 0. \quad (3)$$

Note that the square of a t_n random variable has an $F_{1,n}$ distribution, and if $W \sim F_{m,n}$, then $\frac{1}{W} \sim F_{n,m}$.

CONFIDENCE INTERVALS

Confidence intervals are a way to express the degree of uncertainty around an estimate of a parameter. A confidence interval for a parameter, θ , is a random interval that contains, θ , with some specified probability. When estimated repeatedly, a $100(1 - \alpha)\%$ confidence interval, C.I., will contain the true parameter about $100(1 - \alpha)\%$ times. Most often, we are interested in the 95%, where $\alpha = 0.05$, and 99%, where $\alpha = 0.01$, confidence intervals. The confidence intervals could be one-sided, i.e., of the form $(-\infty, c)$ or (c, ∞) where c is an arbitrary constant, or it could be two-sided, i.e., of the form (c_1, c_2) . Two-sided confidence intervals are usually preferred because they are finite, but one-sided confidence intervals are also frequently used. Confidence intervals could be estimated using various methods. For example, by exact methods, or by approximations based on large sample techniques using asymptotic properties.

3.1 Exact

Suppose that $X_1, X_2, \dots, X_n$ are IID observations from a normal distribution with mean, μ , and variance, σ^2 .

1. If σ^2 is known, then a $100(1 - \alpha)\%$ two-sided C.I. for μ is $\left[\bar{X} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right]$.
2. If σ^2 is unknown, then a $100(1 - \alpha)\%$ two-sided C.I. for μ is $\left[\bar{X} \pm t_{n-1; \frac{\alpha}{2}} \frac{s}{\sqrt{n}} \right]$.

Suppose that the parameter, θ , is such that $\theta = f(\mu)$, where f is a differentiable function.

1. If σ^2 is known, then a $100(1 - \alpha)\%$ two-sided C.I. for θ is $\left[f(\bar{X}) \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} |f'(\bar{X})| \right]$.
2. If σ^2 is unknown, then a $100(1 - \alpha)\%$ two-sided C.I. for θ is $\left[f(\bar{X}) \pm t_{n-1; \frac{\alpha}{2}} \frac{s}{\sqrt{n}} |f'(\bar{X})| \right]$.

3.2 Approximate

When X_i 's are not necessarily normal random variables, whether σ^2 is known or unknown, by the Central Limit Theorem, from approximations based on large samples, the sampling distribution of $\bar{X}$ is approximately Normal. This allows for the use of the same confidence interval framework to be used as above, however we approximate by the standard normal distribution.

3.3 A C.I. based on the MLE

Let $X_1, X_2, \dots, X_n$ be IID with probability function, $f(x; \theta)$. If $f''(x; \theta)$ exists and is continuous, then for large n , an approximate $100(1 - \alpha)\%$ C.I. for θ is given by $\left[\hat{\theta} \pm z_{\frac{\alpha}{2}} \sqrt{\widehat{CRLB}} \right]$, where $\hat{\theta}$ is the MLE for θ , and $\widehat{CRLB}$ is an approximate estimate of the CRLB, $\frac{1}{-\mathbb{E}\left[\frac{\partial^2 L(\theta)}{\partial \theta^2}\right]}$.

CONCLUSION

Confidence intervals provide a range of possible values that an unknown parameter may take. This information is incredible helpful in practice, as it allows us to assess the precision and reliability of statistical estimates.

Whilst this article focuses on interval estimation for general model parameters, the same principles extend naturally to regression models, where confidence intervals can be constructed for regression coefficients. We will examine interval estimation for linear and multilinear regression models soon.

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